

```

#include<iostream>
using namespace std;

void demo() {
    static int count = 0; //static variable..
    cout << count << " ";
    count++;
}
void main()
{
    for (int i = 0; i < 5; i++)
        demo();
}

```

```

#include <iostream>
using namespace std;
int x = 1;
void a() {    int x = 25;  cout << x << endl; cout << ++x << endl;}
void b() {    static int x = 50;
    cout << endl << x << endl; cout << ++x << endl;}
void c() {    cout << endl << x << endl; x *= 10;cout << x << endl;}
void main()
{
    int x = 5;    cout << x << endl;
    {    int x = 7;  cout << x << endl;    }
    cout << x << endl;
    a(); b(); c();
    a(); b(); c();
    cout << x << endl;
}

```

```

#include <iostream>
using namespace std;
void count_1_To_10()
{
    for (int i = 1; i < 11; i++)
        cout << i << endl;
}
void Count_N_to_1(int n) {
    for (int i = n; i >= 1; i--)
        cout << i << endl;
}
void main()
{
    count_1_To_10();
    cout << endl;
    Count_N_to_1(10);
}

```

Overloading

```
#include <iostream>
using namespace std;
int squre(int a)
{
    return a * a;
}
double squre(double a)
{
    return a * a;
}
void main()
{
    cout << "square is: " << squre(3) << endl;
    cout << "square is: " << squre(3.5) << endl;
}
```

التتابع العودية:

```
#include <iostream>
using namespace std;

void printNu(int n) {
    cout << n << endl;
    if (n < 10)
        printNu(++n);
}

void main()
{
    printNu(0);
}
```

$$S=5 + 10 + 15 + 20 + 25 + 30 + .. + 5*N;$$

```
#include <iostream>
using namespace std;

int Sum(int n) {
    if (n == 1)
        return 5;
    else
        return 5 * n + Sum(n - 1);
}

void main()
{
    for (int i = 1; i < 11; i++)
    {
        cout << Sum(i) << endl;
    }
}
```

```
#include <iostream>
using namespace std;
```

```
long Fa(int n) {  
    if (n == 1)  
        return 1;  
    else  
        return n * Fa(n - 1);  
}  
void main()  
{  
    for (int i = 1; i < 6; i++)  
    {  
        cout << Fa(i)<<endl;  
    }  
}
```